SHANXIAO WANG, PhD

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TECHNICAL SKILLS

Domain	Quantitative modeling, credit risk, stress testing, econometrics, macro-finance
Languages	Python, R, SQL, MATLAB, SAS, Stata
Libraries	scikit-learn, statsmodels, pandas, NumPy, Matplotlib
Modeling	Mathematical and statistical modeling, machine learning, DSGE/macroeconomic modeling, classification/regression, model evaluation and validation
Data	Data management, data visualization, dashboards
GenAI	LLM tools and workflows; applied NLP concepts

PROJECTS

Credit Stress-Testing Engine

- Built an end-to-end credit stress-testing pipeline on LendingClub data: a leakage-safe 24-month PD model, a macro satellite model isolating a robust, correctly-signed macroeconomic channel stable across model specifications, and a scenario engine transmitting 2001- and 2008-09-era stress paths into expected-loss estimates.
- Model-risk component: identified and corrected a survivorship-bias flaw in an earlier design via a full diagnostic rebuild, documenting the methodology and findings for model validation.

Macro-Finance PhD Dissertation

- Built a DSGE model with heterogeneous retail and wholesale banks that differ in leverage constraints and balance-sheet behavior. Showed how the expansion of wholesale banking amplifies the transmission of financial risk shocks into greater business-cycle volatility; calibrated to U.S. Flow of Funds data.
- Incorporated financial regulation as a structural model parameter, quantifying how interbank deregulation (e.g., Glass-Steagall repeal) drove the post-1980s growth of wholesale banking.
- Estimated a structural VAR (quarterly U.S. data, 1986–2019) showing that macroeconomic uncertainty transmits through bank leverage and balance sheets to aggregate investment and output, with wholesale banks driving the channel.

Booster Landing Prediction

- Built an end-to-end ML pipeline predicting SpaceX first-stage booster landing outcomes: data collection via API and web scraping, cleaning, feature engineering, EDA with SQL and visualization, model building, model evaluation and validation.
- Built and compared classification models (logistic regression, SVM, decision tree, KNN) to predict first-stage landing success (binary outcome) from launch features. Evaluated models on discrimination and accuracy (ROC-AUC, confusion matrix, cross-validated grid search for hyperparameter tuning).
- Deployed a live dashboard and an interactive map analyzing launch site geography.

Additional projects: shanxiaow.github.io; github.com/shanxiaow

EDUCATION & CERTIFICATION

PhD, Economics (Macro-Finance) 2021

University of Maryland, College Park

- Dissertation: *Essays on Wholesale Banking and the Macro-Economy* — DSGE and structural VAR analysis of risk shocks, bank leverage, and macroeconomic outcomes.
- Graduate training in numerical and computational methods, including optimization, Monte Carlo simulation, and Bayesian estimation.

MS, Epidemiology 2009

Johns Hopkins Bloomberg School of Public Health

- Advanced coursework in statistical analysis, causal inference, and research methodology. Completed thesis-based graduate research.

IBM Data Science Professional Certificate 2023

EXPERIENCE

Faculty, School of Data Science; School of Economics 2022 – 2025

Yokohama City University — Yokohama, Japan

- Designed and delivered courses in Intermediate Macroeconomics, Statistical Analysis, and Data Analysis with R, with sole responsibility for curriculum design, instruction, and evaluation.
- Built applied coursework connecting statistical reasoning, programming, economic modeling, and data analysis for undergraduate students.
- Translated complex quantitative methods into clear lectures, assignments, and feedback for students with varying technical backgrounds.

Instructor, Department of Economics 2014 – 2017

Teaching Assistant, MS in Applied Economics Program 2017 – 2019

University of Maryland, College Park — College Park, MD

- Taught upper-level undergraduate courses in statistical analysis and computational methods in economics.
- Supported graduate coursework in macroeconomics, microeconomics, econometrics, and financial economics.
- Instructed students in Excel, Stata, MATLAB, and Python for applied economic analysis.

Data Analyst, Demographic and Health Surveys (DHS), USAID 2009 – 2013

ICF International — Calverton, MD

- Managed and analyzed nationally representative survey data covering 100+ countries.
- Prepared statistical outputs, visualizations, tables, and graphics for research proposals, manuscripts, presentations, and policy briefs.
- Contributed to capacity-building and project evaluation efforts in developing-country contexts, translating technical analysis into usable documentation.